Adhvaith Vijay

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Product-Focused Data Scientist Turning Models into Decisions

EXPERIENCE:

Senior Data Scientist, Tesla, Palo Alto - Jul 2024 - Present

- Own go/no-go qualification for 10+ vendor and in-house battery cell programs as cell owner
- Built computer vision models that segment 50+ components of every Tesla 4680 battery cell in CT scans
- Validated core collapse detection (the cell's inner core caving in, Tesla's most critical battery defect mode) at 98% recall and 95% precision on in-house cells, clearing it to screen every cell Tesla qualifies
- Forecast thermal-runaway risk (battery overheating) and warranty life with time-series models on 3.5M vehicles' telemetry, informing design decisions with engineering and quality leadership
- Scaled CT inspection to 2,000+ battery cells/day on a distributed Celery/Redis pipeline across 6 VMs
- Shipped quality dashboards that surface battery cell metrology from the defect detection models; 45 engineers across 4 teams use them daily, saving 200+ technician hours/week
- Built a GPT-4o RAG assistant on Databricks Vector Search answering vendor battery cell questions with citations

Data Scientist, Standard AI, San Francisco - Sep 2023 - Jun 2024

- Lifted shopper conversion 29% with randomized in-store A/B tests built on camera-based 3D shopper tracking
- Partnered with product and sales to define engagement metrics behind $600K in new contracts
- Grew revenue 15% across two stores by testing layout changes with Monte Carlo simulations of shopper paths

Cell Engineer Intern, Tesla, Palo Alto - Jan 2023 - Sep 2023

- Eliminated the need for five $27,000 Keyence machines by measuring cell lid bulge via CT within 50 microns
- Cut electrolyte leaks 20% in 5 months with a Tableau dashboard linking dissections to production data

Summer Associate, Credit Fraud Risk, JPMorgan Chase, Wilmington - Jun 2022 - Aug 2022

- Lifted a PySpark XGBoost loan default model 14% with features for erratic repayment and revenue seasonality
- Explained high-risk applicant segments to federal regulators using hierarchical clustering

Undergraduate Researcher, Rolls-Royce North America, Remote - Oct 2019 - May 2020

- Built a pipeline on per-flight engine telemetry to set optimal compressor wash intervals
- Cut aircraft maintenance triage from 3-4 hours to under 5 minutes with an R Shiny tool

EDUCATION:

M.S. in Data Science, University of California, San Diego | Sep 2022 - Mar 2024

- Specialization: Machine Vision and Interaction Design

B.S. in Statistics and Data Science, University of California, Los Angeles | Sep 2019 - Jun 2022

- Dean's Honors List every quarter | Research: UCLA DataRes, CycleGAN image style transfer (Make-A-Monet)

AWARDS AND RECOGNITION:

- Tesla Spot Option Grant for outstanding work, awarded as my team's top performer (Oct 2025)
- 1st place, Chevron track, Rice Datathon (2021)
- 1st place overall, Berkeley DSS Datathon (2020)
- 2nd place, Prosperity track, UCSD DataHacks (2021)
- 2nd place overall, Retina AI R Datathon (2020)

TECHNICAL SKILLS:

- Python, R, SQL, PySpark, PyTorch, scikit-learn, XGBoost, OpenCV, RAG Pipelines, MCP, Snowflake, AWS, Databricks Vector Search, Docker, MLflow, Tableau, A/B Testing

PUBLICATIONS:

- Vijay, A. M., Zanlungo, F., Woollard, D. M., & Morales, L. Y. Analyzing the Shopping Journey: Computing Shelf Visits in a Physical Retail Store. avaitt.github.io/research/2601.00928v1.pdf ***IEEE RO-MAN (under review)***
- Doering, N., Gorlla, C., Tuttle, T., & Vijay, A. (2024) Empirical Analysis of Efficient Fine-Tuning Methods for Large Pre-Trained Language Models. arxiv.org/abs/2401.04051 ***(arXiv 2024)***
- Morales Saiki, L. Y., Vijay, A., et al. (2024) Subject-tracking in a cashier-less store. US2024193497A1 ***(Patent)***